

Cybersecurity lab report

Attacker VM setup and Network Verification.

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Table of Contents.

Contents

1). Introduction.....	3
2). Lab environment.	4
3.1). Hostname.....	5
3.2). Current User.....	6
4). Network Configuration.....	7
4.1). Using ip a	7
4.2). Using ifconfig.....	8
5). Test connectivity with Ping.	9
6). Listening services.	10
7). Conclusion.....	11

1). Introduction.

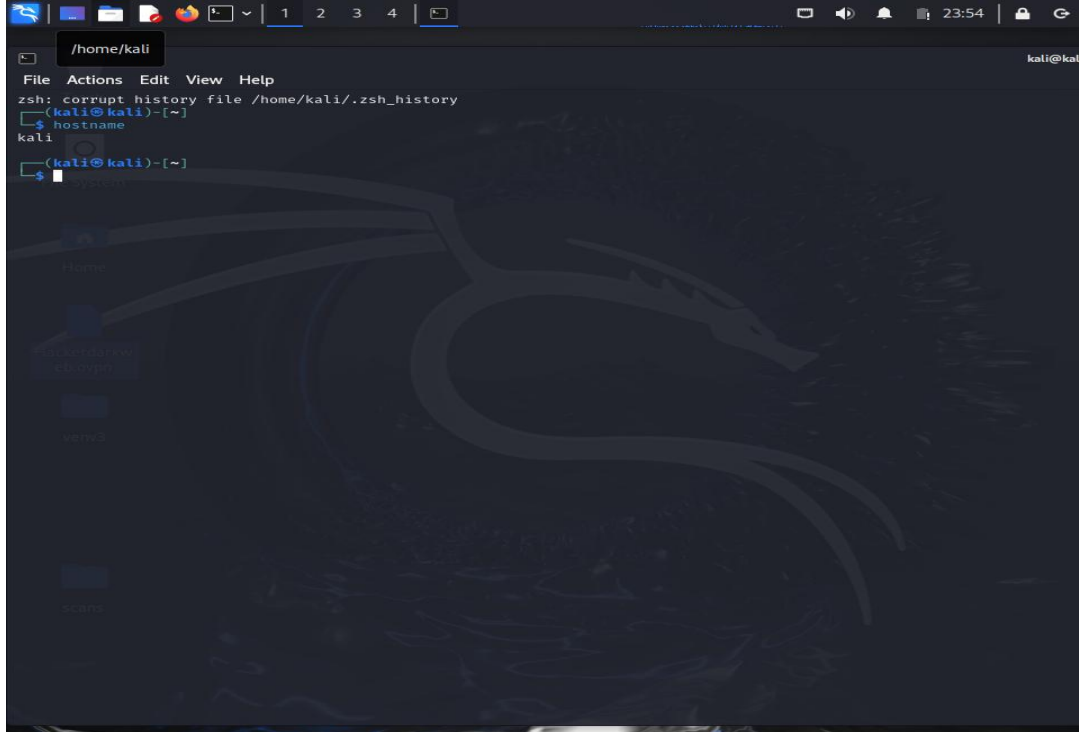
This report documents the setup and verification of a Kali Linux attacker virtual machine using VirtualBox. Several linux networking commands were executed to identify the system, examine network configuration, verify connectivity and identify listening services. Screenshots are included as evidence of each task completed.

2). Lab environment.

Component	Details
Virtualization	Oracle virtualBox
Attacker VM	Kali Linux
Target VM	Metasploitable 2
Network mode	Host-Only Adapter

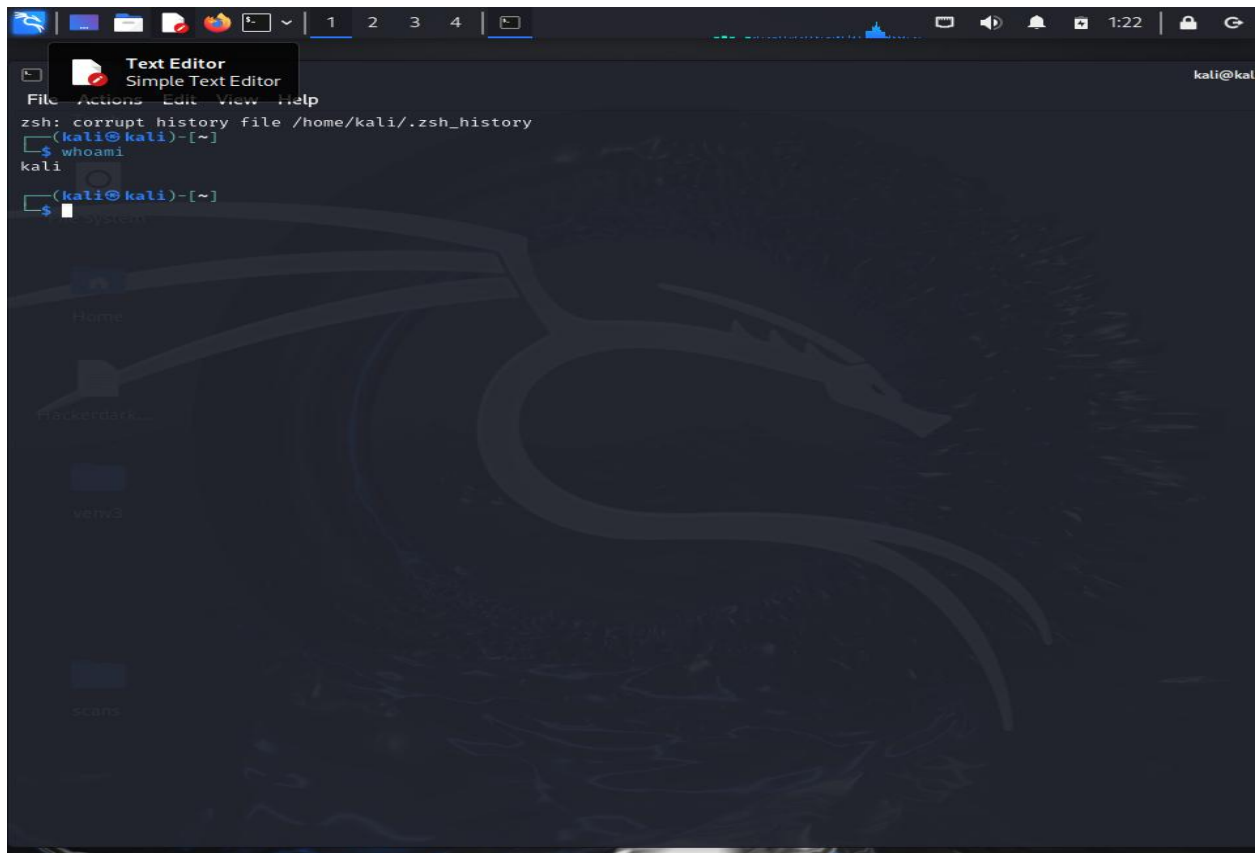
3). System identification.

3.1). Hostname.



The hostname command displays the name assigned to the Kali Linux virtual machine.

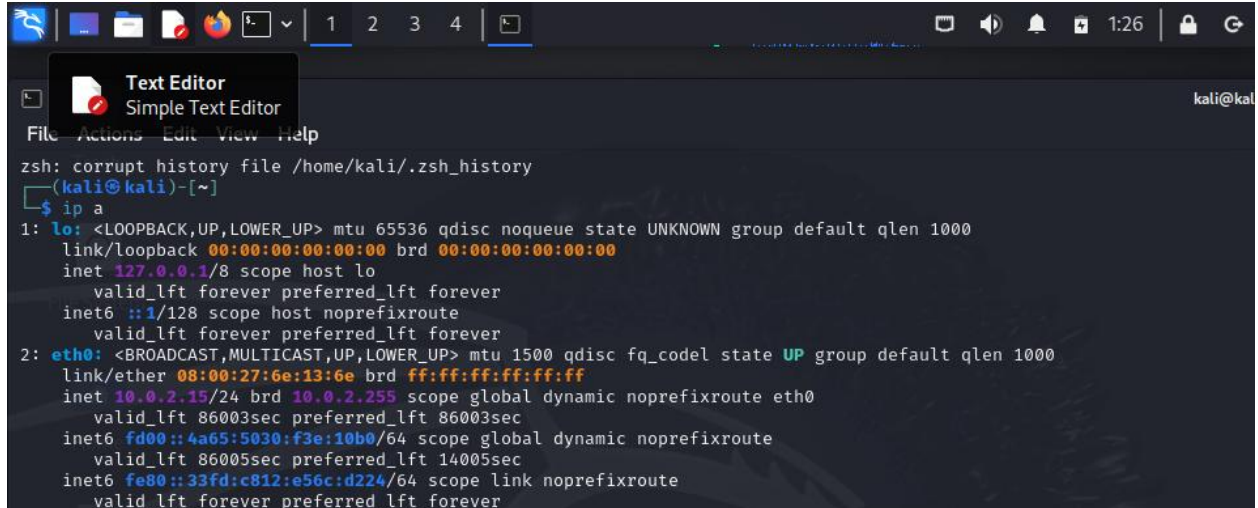
3.2). Current User.



The “whoami” command identifies the currently logged-in user.

4). Network Configuration.

4.1). Using ip a



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:6e:13:6e brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86003sec preferred_lft 86003sec
    inet6 fd00::4a65:5030:f3e:10b0/64 scope global dynamic noprefixroute
        valid_lft 86005sec preferred_lft 14005sec
    inet6 fe80::33fd:c812:e56c:d224/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

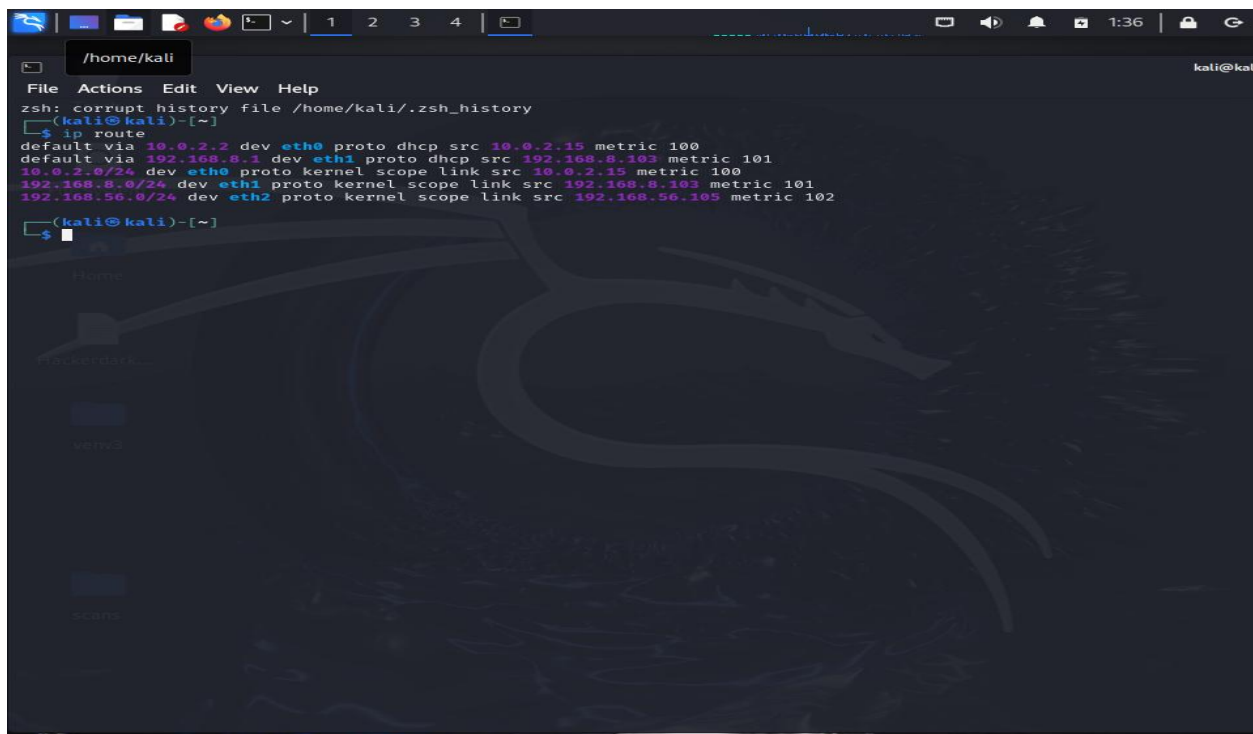
Parameter	Value
Interface	Eth0
IPv4 address	10.0.2.15
Subnet	/24
Status	UP

4.2). Using ifconfig.

```
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd00::4a65:5030:f3e:10b0 prefixlen 64 scopeid 0<global>
    inet6 fe80::33fd:c812:e56c:d224 prefixlen 64 scopeid 0<link>
    ether 08:00:27:6e:13:6e txqueuelen 1000 (Ethernet)
    RX packets 300 bytes 166977 (163.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 276 bytes 36207 (35.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

The ifconfig command displays detailed information about each network interface.

4.3). Default gateway.



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
default via 192.168.8.1 dev eth1 proto dhcp src 192.168.8.103 metric 101
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
192.168.8.0/24 dev eth1 proto kernel scope link src 192.168.8.103 metric 101
192.168.56.0/24 dev eth2 proto kernel scope link src 192.168.56.105 metric 102

(kali@kali)-[~]
$
```

Parameter	Value
Default gateway	10.0.2.2
Preferred interface	Eth0
Metric	100

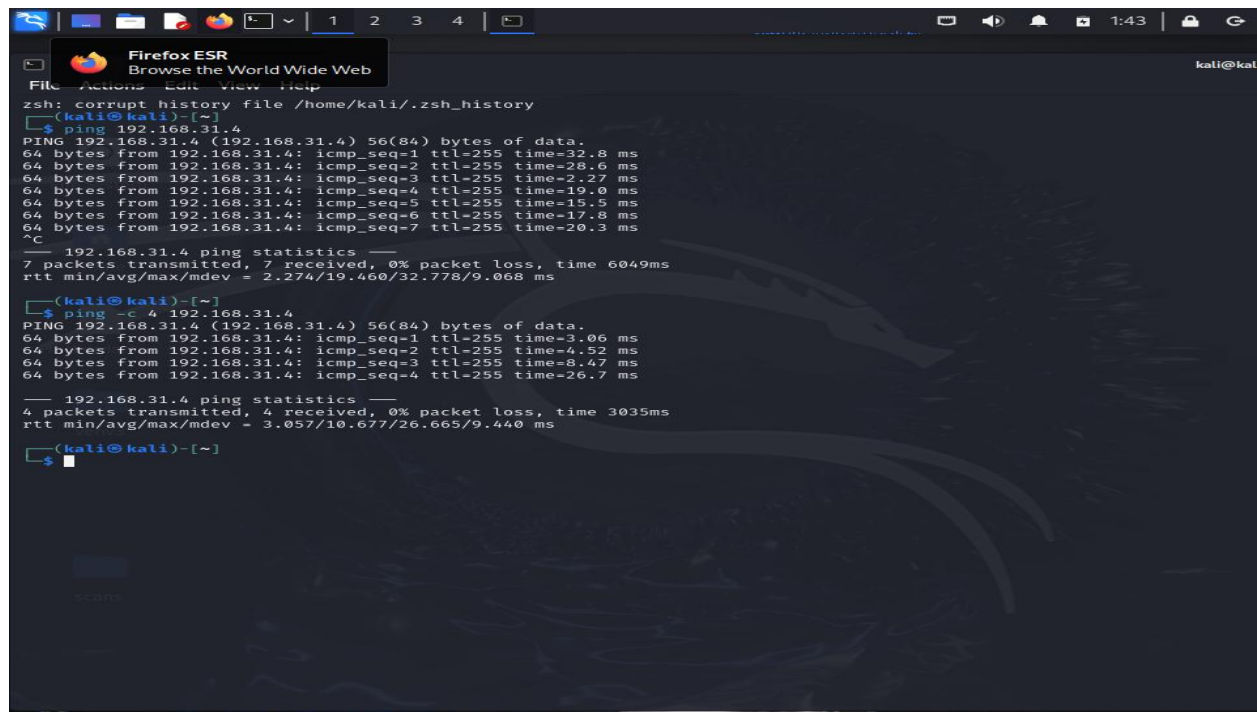
The ip route command was used to identify the routing configuration of the Kali Linux system. The default route indicates the gateway used by the attacker machine to reach networks outside its local subnet.

5). Test connectivity with Ping.

```
link/ether 08:00:27:d0:ff:2f brd ff:ff:ff:ff:ff:ff
inet 192.168.31.4/24 brd 192.168.31.255 scope global eth0
inet6 fe80::a00:27ff:fed0:ff2f/64 scope link
    valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:d0:ff:2f
          inet addr:192.168.31.4  Bcast:192.168.31.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fed0:ff2f/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:59 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:3720 (3.6 KB)  TX bytes:9118 (8.9 KB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128  Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:142 errors:0 dropped:0 overruns:0 frame:0
          TX packets:142 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:42429 (41.4 KB)  TX bytes:42429 (41.4 KB)
msfadmin@metasploitable:~$ _
```

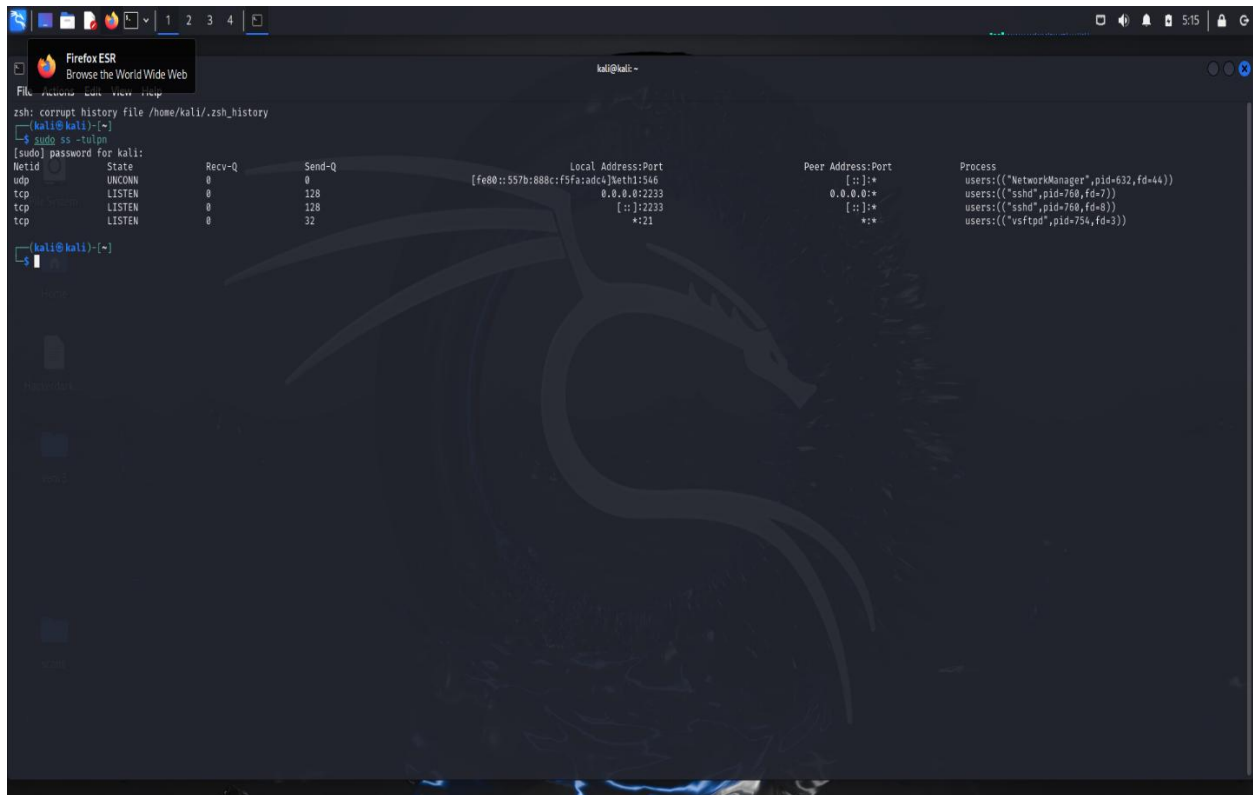
This is the ip address of the metasploitable VM.



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ ping 192.168.31.4
PING 192.168.31.4 (192.168.31.4) 56(84) bytes of data:
64 bytes from 192.168.31.4: icmp_seq=1 ttl=255 time=32.8 ms
64 bytes from 192.168.31.4: icmp_seq=2 ttl=255 time=28.6 ms
64 bytes from 192.168.31.4: icmp_seq=3 ttl=255 time=2.27 ms
64 bytes from 192.168.31.4: icmp_seq=4 ttl=255 time=19.0 ms
64 bytes from 192.168.31.4: icmp_seq=5 ttl=255 time=15.5 ms
64 bytes from 192.168.31.4: icmp_seq=6 ttl=255 time=17.8 ms
64 bytes from 192.168.31.4: icmp_seq=7 ttl=255 time=20.3 ms
^C
--- 192.168.31.4 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6049ms
rtt min/avg/max/mdev = 2.274/19.460/32.778/9.068 ms
(kali@kali)~$ ping -c 4 192.168.31.4
PING 192.168.31.4 (192.168.31.4) 56(84) bytes of data:
64 bytes from 192.168.31.4: icmp_seq=1 ttl=255 time=3.06 ms
64 bytes from 192.168.31.4: icmp_seq=2 ttl=255 time=4.52 ms
64 bytes from 192.168.31.4: icmp_seq=3 ttl=255 time=8.47 ms
64 bytes from 192.168.31.4: icmp_seq=4 ttl=255 time=26.7 ms
^C
--- 192.168.31.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3035ms
rtt min/avg/max/mdev = 3.057/10.677/26.665/9.440 ms
(kali@kali)~$
```

A four-packet ICMP ping test was performed from the Kali Linux attacker machine to the laboratory target VM. The successful responses demonstrate network connectivity between the two virtual machines.

6). Listening services.



```
kali@kali:~$ sudo ss -tulpn
zsh: corrupt history file /home/kali/.zsh_history
[kali@kali]~$ sudo ss -tulpn
[sudo] password for kali:
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
udp UNCONN 0 0 *:*:* users:((("NetworkManager",pid=632,fd=44))
tcp LISTEN 0 128 0.0.0.0:2233 0.0.0.0:* users:((("sshd",pid=760,fd=7))
tcp LISTEN 0 128 :::2233 :::* users:((("sshd",pid=760,fd=8))
tcp LISTEN 0 32 *:*21 *:* users:((("vsftpd",pid=754,fd=3))
```

The sudo is used to get into root mode to perform listening services to see which ports are active. The ss command was used to identify network sockets and services currently listening on the kali Linux system. The -t option displays TCP sockets, -u displays UDP sockets, -l shows listening sockets, -p displays the associated processes and -n displays numerical addresses and port numbers. Results indicate that the Kali Linux SSH (sshd) listening on TCP port 2233 and FTP (vsftpd) listening on TCP port 21.

7). Conclusion.

The kali linux attacker environment was successfully configured and verified. System identification was performed using the hostname and whoami commands, while network configuration was examined using ip a and ifconfig. The routing configuration and default gateway were identified using Ip route. Network connectivity between the attacker and laboratory environment was verified using ICMP ping testing. Finally, listening network services were identified using the ss command. The collected screenshots provide evidence that the attacker VM is correctly configured and ready for subsequent authorized security testing activities.